

Math 514, F19
Quiz 12

Name: Answer

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (10 points) The prices of a security follow a geometric Brownian motion with drift parameter $\mu = 0.05$ and volatility parameter $\sigma = 0.4$. The security's present price is 200. A new investment that is being marketed costs 20; after 2 year the investment will pay 25 if $S(2) < 180$, will pay x if $S(2) > 210$, and will pay 0 otherwise. The nominal interest rate is 6%, continuously compounded.

(a) What must be the value of x if this new investment is not to give rise to an arbitrage?

$$t=2 \quad (r - \sigma^2/2)t = (0.06 - 0.4^2 \times 0.5) \times 2 = -0.04$$

$$\sigma\sqrt{t} = 0.4 \times \sqrt{2} \approx 0.5656$$

Let W be the normal random variable with mean -0.04 and standard deviation 0.5656 . We then have $S(t) = S e^{W(t)}$

$$20 = (25 P\{S(2) < 180\} + x P\{S(2) > 210\}) e^{-0.06 \times 2}$$

$$e^{0.12} \times 20 = 25 P\{W < \log(\frac{180}{200})\} + x P\{W > \log(\frac{210}{200})\}$$

$$e^{0.12} \times 20 = 25 P\{Z < \frac{\log(0.9) + 0.04}{0.5656}\} + x P\{Z > \frac{\log(1.05) + 0.04}{0.5656}\}$$

$$e^{0.12} \times 20 = 25 \Phi(-0.1157) + x(1 - \Phi(0.1572)) \Rightarrow x = \frac{20e^{0.12} - 25\Phi(-0.1157)}{1 - \Phi(0.1572)}$$

(b) What is the probability that $S(1) > 180$?

$$\approx 25.6$$

$$\mu t = 0.05 \times 1 = 0.05$$

$$t=1 \quad \sigma\sqrt{t} = 0.4 \times \sqrt{1} = 0.4$$

Let W be the normal random variable with mean 0.05 and standard deviation 0.4 . Let $S(t) = S e^{W(t)}$

$$P\{S(1) > 180\} = P\{W > \log(\frac{180}{200})\} \\ = P\{Z > \frac{\log(0.9) - 0.05}{0.4}\}$$

$$= 1 - \Phi(-0.2304)$$

$$\approx 0.5911$$